

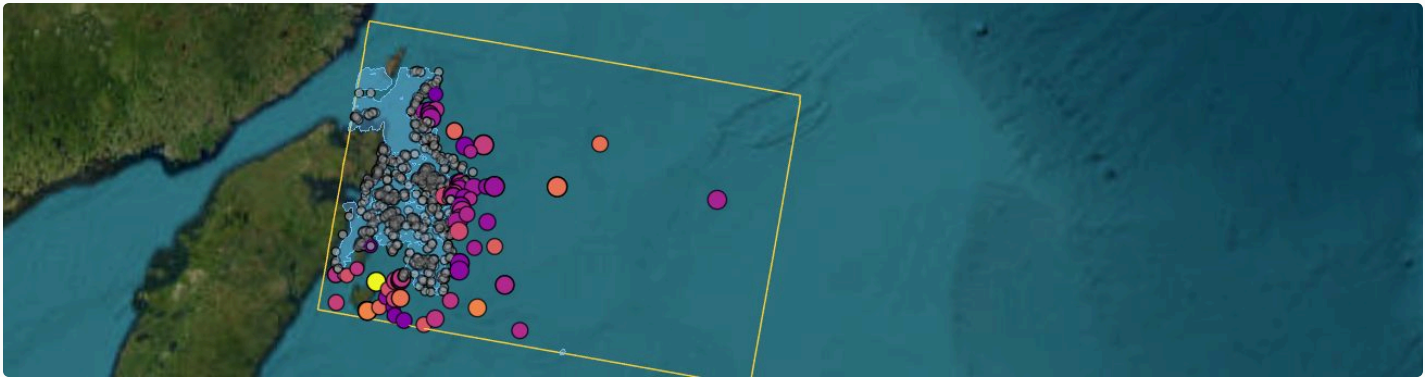
Iceberg Alley SAR

Detecting icebergs off Newfoundland in Sentinel-1 satellite radar, validated against International Ice Patrol data and shown on an interactive 3D globe.

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Independent project, 2026

github.com/natali-kovalski/iceberg-alley-sar



The viewer, 2 May 2025: scene footprint (yellow), pack ice (blue), candidates coloured by radar contrast, likely ice floes in grey.

78% recall against Ice Patrol analyst labels on the same pass (209 of 268)	128 m median position difference to the labelled bergs	6 Sentinel-1 scenes processed at 10 m (2019, 2020, 2025)	1 command from scene search to globe, covered by 67 automated tests
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The problem

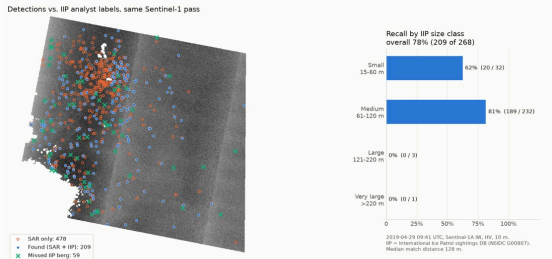
Each spring, icebergs from Greenland drift down the Labrador Current past northeast Newfoundland, into shipping lanes, fishing grounds and offshore platform areas. Aircraft patrols are costly and grounded by fog. Synthetic aperture radar (SAR) images the ocean through cloud and darkness. The question: how reliably can free Sentinel-1 data find these bergs?

Approach

- Search and order.** Find HH/HV scenes over the area with ASF and order calibrated, terrain-corrected products (Hyp3 RTC, 10 m).
- Mask.** Land from OpenStreetMap plus a 500 m buffer; pack ice from cross-pol (HV) texture on 200 m cells.
- Detect.** Cell-averaging CFAR on HV intensity. The threshold follows from a 10^{-6} false-alarm rate and speckle statistics measured in each scene.
- Filter.** Size limits, rejection of thin sea-ice strips, and an HH co-polarization check that removes speckle at 10 m.
- Publish.** GeoJSON with per-target attributes, then time-dynamic CZML for a Vue + CesiumJS viewer.

Validation

Ground truth is scarce, so I tested four sources: ice charts, aircraft sightings, Sentinel-2 optical imagery, and Ice Patrol analyst labels on the same radar pass. Only the last supports a per-berg score. The others were limited by coarse resolution, hours of drift, or sea ice that optical imagery cannot tell from bergs. The code reports those cases as unresolved instead of forcing a score.



29 April 2019: detections vs. analyst labels. Recall is 62% for small and 81% for medium bergs; the four largest were missed.

Engineering practice

- Modular Python package** with type hints and a CLI
- One config file**, lengths in metres so tuning holds across resolutions
- Re-runnable pipeline** that reuses data and shows processing cost first
- Findings and limits** written up next to the code

Limits and next steps

Precision needs AIS ship positions to measure, and bergs over 120 m are missed. Next: AIS matching, size-aware filters, and OS-CFAR for heavy-tailed sea clutter.

TOOLS

Python · NumPy · SciPy · rasterio · GeoPandas · Hyp3 · Vue 3 · CesiumJS